

Tareq Shawabkeh

AI, Machine Learning, Computer Vision & Data Engineer

tareqshawabkehwork@gmail.com | +962 (0) 786415882 | [LinkedIn](#) | [GitHub](#) | Amman, Jordan

SUMMARY

Computer Engineering Fres graduate specializing in Artificial Intelligence, Machine Learning, and Data Engineering. with strong hands-on experience in building real-time, AI-integrated systems and API-driven applications. Experienced in developing production-style systems from design to deployment, including scalable backend architectures (Fast API), secure data validation pipelines, and enterprise-grade REST APIs. Proven ability to deliver award-winning solutions, highlighted by engineering an AI-automated trading platform (Graduation Project 4/4) that won First Place at the NTP2025 National Competition. Strong in machine learning workflows, system design, and transforming data into reliable, production-oriented solutions. Seeking an opportunity to grow in AI and data engineering within a practical, real-world environment. with practical experience gained through internships and academic teaching assistance.

EDUCATION

B.Sc. in Computer Engineering University of Jordan — Amman, Jordan | Graduated: January 2026
Graduation Project: “Real-Time and AI-Based Automated Trading App” (ML Bot + Backend API + Flutter App).
Relevant Coursework: Python, Machine Learning, Data Structures, Network Programming, Java, Probability & Statistics.

SKILLS

Programming Languages: *Python*, Java, C++, SQL, Dart-Flutter
Backend & Systems: FastAPI, REST APIs, WebSocket’s, Multithreading, TCP/IP, Async Processing, Client–Server Architecture, Secure API Design & Data Validation.
Databases: *PostgreSQL*, *SQLite*, SQLAlchemy, JPA/Hibernate.
AI / Data: Scikit-learn, TensorFlow, data processing with Pandas, NumPy, Data preprocessing & feature engineering.
Tools & Practices: Docker, Postman, Swagger/OpenAPI, Git, VS Code, NetBeans, API Design, Real-Time Trading systems, Secure Key Management, Client–Server Architecture.

ACHIEVEMENTS

- 🏆 First Place Winner — NTP2025 Competition (Jordan)
- Top 10 Graduation Projects in School of Engineering & IT — University of Jordan

PROJECTS

- AI-based Automated Trading System**
🏆 **First place winner NTP2025 - Graduation Project.** *University of Jordan | Oct 2024 – May 2025*
Designed and implemented a full-stack, real-time automated trading system for cryptocurrency markets, combining machine learning models, trading strategy, with a high-performance backend - ensuring multi-user support, encrypted API key handling and live execution infrastructure (tracks performance in real time).
The system processes live market data, executes strategies and custom ML models were built from scratch that achieved 161% profit ratio vs only 8.44% loss in long back-testing.
Key Contributions:
 - Architected a modular system integrating a Flutter mobile app, FastAPI backend, SQLite database, and broker APIs for end-to-end automation
 - Developed a real-time trading engine that continuously processes live market data, generates signals, and executes trades automatically
 - Designed an LSTM-based machine learning model for time-series prediction, evolving from price regression to feature-based classification using indicators (MACD, RSI)
 - Designed and implemented the backend using FastAPI with async processing and thread-based trading loops.
 - Built secure multi-user architecture with encrypted API key storage and session persistence.
 - Integrated multiple trading strategies (SMA, MACD, RSI, ADX + ML - based model) with real-time execution.
 - Implemented WebSocket-based live updates for trades, balance, and performance tracking.
 - Collaborated on Flutter mobile app for strategy control and monitoring.**Tech Stack:** FastAPI, SQLite, SQLAlchemy (Async), WebSocket, Threading, TensorFlow, Cryptography, Flutter.
- Car Price Prediction using Advanced Regression Techniques (AI Course)** *Jan 2024 – Mar 2024*
 - Built an end-to-end car price prediction pipeline starting from raw CSV data to model evaluation.
 - Designed a CustomTransformer preprocessing pipeline to handle mixed features: numerical attributes were scaled, while categorical data (make, model, color, state) was encoded in a structured and reusable way.
 - Used the same preprocessing pipeline across training, validation, and testing to ensure consistent data flow and prevent data leakage.
 - Trained and compared multiple models, then extended the work to neural networks and deep learning architectures, including CNN-based feature learning.
 - Evaluated model performance using RMSE and validation loss to select stable and generalizable configurations.
- FX Deals Importer System (Spring Boot, PostgreSQL)**
 - Built REST APIs for real-time FX deal submission and bulk CSV import.
 - Designed layered architecture (Controller, Service, Repository, DTO, Entity) to ensure clean separation of concerns.
 - Documented APIs using Swagger and tested endpoints using Postman.
 - Dockerized the application for consistent deployment.

4. Client–Server Medical Consultation Platform (Java Backend, TCP/IP, Multithreading)

Course Project – Network Programming, Spring 2024
- Developed a multithreaded Java Backend Application simulating a real-time medical consultation system. The server handled two user roles-doctors (with secure login, registration, and specialty-based ticket queues) and patients (submitting general or directed medical inquiries).
- Implemented TCP/IP-based client–server communication.
 - Designed concurrent server handling for multiple doctors and patients.
 - Built secure login, ticketing, and queue management system.
 - Doctor registration/login and real-time ticket queue management (e.g., 5 patients per doctor)
 - Patient ability to browse online doctors by specialty and submit general or direct tickets
 - Implemented FIFO scheduling, timeout handling, and availability checks.

5. Aerial Image Object Detection Pipeline (Computer Vision)

2026 Apr
- Developed** an end-to-end computer vision pipeline using YOLOv8 Nano to accurately detect and track vehicles in high-resolution, top-down aerial imagery.
 - Engineered** a robust data ingestion module that dynamically parses PASCAL VOC XML annotations to YOLO format while enforcing deterministic dataset splitting.
 - Achieved** an outstanding model performance of 99.0% mAP@50, specifically optimizing the architecture for lightweight, edge-computing environments like drones.
 - Designed** a flexible Command-Line Interface (CLI) using argparse to facilitate seamless, path-independent inference on both test sets and unseen "wild" data.
 - Ensured** strict cross-platform reproducibility by building a pinned, CPU-optimized Python environment, completely eliminating dependency and architecture conflicts.

EXPERIENCE

- Software Engineering Intern - Link Business Solutions Company.

Amman, Khalda, Jordan | Summer 2025
- Contributed to web system implementation using HTML, CSS, JavaScript, and Java within a professional development environment.
 - Participated in transforming UI designs into functional web pages and interactive while following company standards and clean coding practices (e.g., TO-DO-List).
 - Worked with backend developers to integrate frontend components with server-side logic (e.g., **The FX deals Backend Project**).*
 - Gained exposure to real production workflows, requirement changes, and team-based delivery.

- Teaching Assistant - Python Programming Lab

University of Jordan | 2025 – 2026
- Assisted undergraduate students in Python labs by explaining programming concepts, debugging code, and guiding problem-solving approaches.
 - Supported students during lab sessions and assignments, helping them understand logic, syntax, and program structure.
 - Developed **strong communication** and **teaching skills** by simplifying complex programming ideas for different learning levels.
 - Provided feedback on student code and corrected logical and runtime errors.